

Empirical Realization, Equivalence, and Limits

A Typed Account for Finite-Dimensional Classical Theories

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Abstract

We formulate the distinction between a mathematically admissible model and an empirically realized physical model. An empirical theory is represented by a typed tuple containing models, experimental contexts, preparation and measurement kernels, probability laws, tolerances, and a domain of intended application. Relative to this tuple we define exact and tolerance-indexed observational equivalence. We prove that a mathematical map preserves empirical content only when it intertwines preparation, dynamics, and measurement on the tested domain. We also prove a uniform error-transfer theorem for limiting theories and exhibit failures caused by nonuniform convergence, inaccessible preparations, unmodelled interventions, and structural mismatch. The framework yields finite falsifiers: registered experiments whose discrepancy exceeds a prespecified error budget. The founding thesis that physics studies empirically realized compositional mathematical structures is treated as a conjectural methodological proposal, not as an established physical result.

1 Scope and epistemic status

The analysis concerns finite-dimensional smooth classical systems. Field theory, general relativity, quantum theory, and infinite-dimensional analysis occur only as boundary markers. No new experimental finding is asserted.

Conjecture 1.1 (Founding thesis). *Physics is the study of empirically realized compositional mathematical structures.*

This sentence is not used as an axiom below. The mathematical results establish conditional relations among explicitly defined objects; whether those objects adequately represent a laboratory system remains answerable to data.

Established physical result 1.2 (Hamiltonian prediction). *For a finite-dimensional symplectic manifold (X, ω) and smooth Hamiltonian H , a Hamiltonian trajectory satisfies*

$$\iota_{X_H}\omega = dH, \quad \dot{x} = X_H(x),$$

where existence and uniqueness hold locally because X_H is a smooth vector field. This is standard finite-dimensional Hamiltonian mechanics [1, 2].

The displayed equations define predictions only after specifying which physical preparations correspond to points or distributions on X , which instruments correspond to observables, and which error model applies. The separation is central: differential geometry supplies admissibility; experiment supplies realization.

2 Typed empirical theories

Definition 2.1 (Mathematical model). *A deterministic mathematical model is a tuple*

$$M = (X, U, \Phi, Y),$$

where X is a finite-dimensional smooth state space, U is a set of admissible controls, $\Phi_t^u : X \rightarrow X$ is the evolution where defined, and Y is a measurable outcome space. The model is mathematically admissible relative to a specification Σ when it satisfies the regularity, composition, and dynamical conditions listed in Σ .

Admissibility is relative to a mathematical specification. It neither states that a state can be prepared nor that an outcome is measured by an actual apparatus.

Definition 2.2 (Empirical interpretation). *For a model M , an empirical interpretation on a domain D is*

$$I = (\mathcal{C}, \mathcal{P}, \mathcal{Q}, K, \epsilon, D).$$

Here $\mathcal{C}$ is a set of registered experimental contexts; $\mathcal{P}_c$ is a set of physical preparation procedures; $\mathcal{Q}_c$ is a set of measurement procedures; $K_c^P(dx)$ is a preparation kernel on X ; $K_c^Q(dy | x)$ is a measurement kernel on Y ; $\epsilon(c) > 0$ is a declared total error tolerance; and $D \subseteq \mathcal{C}$ is the intended domain. With control u_c and duration t_c , the predicted outcome law is

$$\Pr_{M,I}(A | c) = \int_X K_c^Q(A | \Phi_{t_c}^{u_c}(x)) K_c^P(dx). \quad (1)$$

Kernels allow noisy preparation and measurement. Deterministic readout $h : X \rightarrow Y$ is recovered by $K^Q(A | x) = \mathbf{1}_A(h(x))$.

Definition 2.3 (Empirical theory object). *An empirical theory object is $T = (M, I, \mathcal{E})$, where M is a mathematical model, I is an empirical interpretation, and $\mathcal{E}$ records calibration evidence, uncertainty estimates, and provenance. A physical target S realizes T on D to tolerance ϵ when, for every registered $c \in D$, the target distribution P_S^c exists and*

$$d_{\text{TV}}(P_S^c, \Pr_{M,I}(\cdot | c)) \leq \epsilon(c). \quad (2)$$

For continuous outcomes another prespecified metric may replace total variation; changing the metric changes the claim.

Empirical statement 2.4 (Realization input). *Any assertion that a concrete target realizes T requires observed data, calibration records, and a justified statistical procedure. Equation (2) is a criterion, not evidence that any target satisfies it.*

Interpretation 2.5. *The evidence component prevents an interpretation map from functioning as an empirical certificate merely because it was written down. Realization is indexed by domain, tolerance, intervention family, and date-sensitive calibration state.*

Counterexample 2.6 (Admissibility without realization). *Let $X = \mathbb{R}^2$, $\omega = dq \wedge dp$, and $H(q, p) = -(q^2 + p^2)/2$. This is a smooth Hamiltonian system with complete circular trajectories. If the intended target is a free positive-mass particle, its measured motion is not circular and no ordinary free-particle preparation/measurement interpretation makes the model accurate. Smooth symplectic admissibility therefore does not entail realization by that target.*

3 Observational and physical equivalence

Definition 3.1 (Indexed observational equivalence). *Let $T_i = (M_i, I_i, \mathcal{E}_i)$ predict probability laws P_i^c on a common registered outcome space for $c \in D$. Define*

$$\Delta_D(T_1, T_2) = \sup_{c \in D} d_{\text{TV}}(P_1^c, P_2^c).$$

The theories are exactly observationally equivalent on D when $\Delta_D = 0$, and η -observationally equivalent when $\Delta_D \leq \eta$.

This definition is deliberately indexed. Agreement on yesterday's instruments need not survive a new intervention. Empirical equivalence is commonly treated as necessary, but not generally sufficient, for theoretical equivalence; formal equivalences must also preserve empirical content [3, 4].

Lemma 3.2 (Restriction monotonicity). *If $D' \subseteq D$, then $\Delta_{D'}(T_1, T_2) \leq \Delta_D(T_1, T_2)$.*

Proof. The supremum over the subset D' cannot exceed the supremum over D . $\square$

Corollary 3.3 (Underdetermination by restricted experiments). *Observational equivalence on D' supplies no bound on $\Delta_{D \setminus D'}$.*

Proof. Take two Bernoulli theories that both assign probability $1/2$ for every $c \in D'$, while for one context $c_* \in D \setminus D'$ they assign probabilities 0 and 1. Then $\Delta_{D'} = 0$ and $\Delta_D = 1$. $\square$

Definition 3.4 (Structural equivalence). *Two models are structurally equivalent relative to selected structure when there are inverse maps $F : X_1 \rightarrow X_2$ and $G : X_2 \rightarrow X_1$ preserving that selected structure. In Hamiltonian mechanics a strong choice requires a symplectomorphism F and*

$$F \circ \Phi_{1,t}^u = \Phi_{2,t}^{\bar{u}} \circ F.$$

The phrase “selected structure” is indispensable: bijection, diffeomorphism, symplectomorphism, and categorical equivalence preserve different data.

Theorem 3.5 (Empirical preservation by intertwining). *Suppose $F : X_1 \rightarrow X_2$ is measurable and, for every $c \in D$,*

$$K_c^{P,2} = F_* K_c^{P,1}, \tag{3}$$

$$F \circ \Phi_{1,t_c}^{u_c} = \Phi_{2,t_c}^{\bar{u}_c} \circ F \quad K_c^{P,1}\text{-almost surely}, \tag{4}$$

$$K_c^{Q,2}(A \mid F(x)) = K_c^{Q,1}(A \mid x) \tag{5}$$

for all measurable A . Then T_1 and T_2 are exactly observationally equivalent on D .

Proof. For a measurable A , substitute (3) into the prediction for T_2 and use the pushforward integration formula:

$$P_2^c(A) = \int_{X_1} K_c^{Q,2}(A \mid \Phi_{2,t_c}^{\bar{u}_c}(F(x))) K_c^{P,1}(dx).$$

Equation (4) replaces the evolved argument by $F(\Phi_{1,t_c}^{u_c}(x))$. Equation (5) then makes the integrand $K_c^{Q,1}(A \mid \Phi_{1,t_c}^{u_c}(x))$. The resulting integral is $P_1^c(A)$. Equality for every measurable A gives $P_1^c = P_2^c$ for every $c \in D$, hence $\Delta_D = 0$. $\square$

Counterexample 3.6 (Isomorphic dynamics, inequivalent measurements). *Let both state spaces be $\mathbb{R}$, both evolutions the identity, and $F(x) = x$. Prepare $x = 1$. Let $h_1(x) = x$ and $h_2(x) = 2x$ with noiseless readout. The dynamics are isomorphic, but the outcome laws are point masses at 1 and 2, whose total-variation distance is 1. Condition (5) fails.*

Counterexample 3.7 (Same equations, different boundaries). *The differential equation $\ddot{q}+q=0$ on a time interval does not determine an empirical model until initial or boundary data and their physical control are fixed. The families $q(0)=0, \dot{q}(0)=1$ and $q(0)=1, \dot{q}(0)=0$ predict $\sin t$ and $\cos t$. Equality of equations of motion does not ensure observational equivalence. Boundary structure can carry empirical content [5].*

Definition 3.8 (Physical equivalence). *Two empirical theories are physically equivalent on D only if (i) they are observationally equivalent on D , (ii) a correspondence identifies their preparations, interventions, outcomes, and target referents, and (iii) no claimed physically significant structure is lost under that correspondence. This is a criterion adopted here, not an established physical result.*

Proposition 3.9 (Formal equivalence is insufficient). *Structural equivalence of mathematical models alone does not imply physical equivalence.*

Proof. The preceding measurement counterexample has identical state spaces and dynamics, hence the identity is a structural isomorphism, while observational equivalence fails. Since clause (i) in the definition of physical equivalence fails, physical equivalence fails. $\square$

4 Composition and empirical interfaces

Assumption 4.1 (Independent product composition). *For independently prepared systems A and B , the composite state space is $X_A \times X_B$, the preparation kernel factors, and uncoupled dynamics factor:*

$$K_{AB}^P = K_A^P \otimes K_B^P, \quad \Phi_{AB,t} = \Phi_{A,t} \times \Phi_{B,t}.$$

This assumption is not a theorem about every physical composite. Correlated preparation, interaction constraints, and shared environments violate factorization.

Proposition 4.2 (Compositional error bound). *Under independent product composition, if component outcome laws satisfy $d_{\text{TV}}(P_A, Q_A) \leq \epsilon_A$ and $d_{\text{TV}}(P_B, Q_B) \leq \epsilon_B$, then*

$$d_{\text{TV}}(P_A \otimes P_B, Q_A \otimes Q_B) \leq \epsilon_A + \epsilon_B.$$

Proof. Insert the intermediate product $Q_A \otimes P_B$ and use the triangle inequality. Tensoring both measures with the same probability measure preserves total-variation distance, so the two terms are respectively bounded by ϵ_A and ϵ_B . $\square$

Interpretation 4.3. *A mathematical tensor or Cartesian product becomes an empirical claim only when the interface specifies which joint preparations and measurements are available. Error budgets compose, but the stated bound can become uninformative for many components unless correlations and scaling are controlled.*

5 Limits, approximations, and recovery

Definition 5.1 (Uniform empirical limiting relation). *Let T_λ be a family of theories and T_0 a candidate limiting theory on D . We say $T_\lambda \rightarrow T_0$ uniformly empirically on D if*

$$r(\lambda) := \sup_{c \in D} d_{\text{TV}}(P_\lambda^c, P_0^c) \rightarrow 0.$$

Pointwise empirical convergence replaces the supremum statement with convergence for each fixed c .

New result claimed by this work 5.2 (Uniform error-transfer theorem). *Assume the target S is realized by T_λ with error $e_\lambda(c)$, meaning $d_{\text{TV}}(P_S^c, P_\lambda^c) \leq e_\lambda(c)$. If $T_\lambda \rightarrow T_0$ uniformly with error $r(\lambda)$, then*

$$\sup_{c \in D} d_{\text{TV}}(P_S^c, P_0^c) \leq \sup_{c \in D} e_\lambda(c) + r(\lambda). \quad (6)$$

Thus T_0 is empirically adequate to tolerance δ whenever the right-hand side is at most δ .

Proof. For each c , the triangle inequality gives

$$d_{\text{TV}}(P_S^c, P_0^c) \leq d_{\text{TV}}(P_S^c, P_\lambda^c) + d_{\text{TV}}(P_\lambda^c, P_0^c) \leq e_\lambda(c) + r(\lambda).$$

Taking the supremum over D proves (6). $\square$

The theorem is elementary but records the missing empirical premise in many limit claims: convergence among mathematical predictions transfers adequacy only when some family member is already connected to the target by evidence.

Counterexample 5.3 (Pointwise convergence is not uniform). *On $D = [0, 1]$, let a deterministic prediction be $f_n(c) = c^n$ and the proposed limit be $f(c) = 0$ for $c < 1$, $f(1) = 1$. Then $f_n(c) \rightarrow f(c)$ for every fixed c . Yet*

$$\sup_{0 \leq c < 1} |f_n(c) - f(c)| = 1$$

for every n . No single finite n controls all contexts approaching the boundary. A claim of uniform empirical recovery would therefore be false.

Counterexample 5.4 (Noncommuting limit and long observation). *Let $q_\varepsilon(t) = \sin(\varepsilon t)$. For each fixed finite t , $q_\varepsilon(t) \rightarrow 0$ as $\varepsilon \rightarrow 0$. But at the context $t_\varepsilon = \pi/(2\varepsilon)$, $q_\varepsilon(t_\varepsilon) = 1$. Hence convergence on every fixed compact time interval does not license recovery over times scaling as $1/\varepsilon$.*

Definition 5.5 (Approximation versus idealization). *Following Norton’s distinction, an approximation is an inexact description of a target, whereas an idealization is a distinct system whose properties provide an inexact description of the target [6]. We call a limit representationally controlled only when the target, metric, domain, and error rate are stated.*

Infinite or singular limit objects need not be physically realized. Work on limiting relations emphasizes that novel limiting behavior can coexist with finite, physically realized precursor behavior [7, 8]. In the present finite-dimensional scope, this literature functions as a warning against reifying a limit point.

Proposition 5.6 (Limit realization does not follow from convergence). *The statement $T_\lambda \rightarrow T_0$ in a mathematical topology does not by itself imply that a physical target realizes T_0 .*

Proof. Choose any convergent family of mathematical predictions and a target whose observed distribution differs from the limit by total-variation distance 1. Convergence relates members of the model family to the limit; it contains no premise relating the target distribution to any member. Equation (6) identifies the absent premise. $\square$

6 Failure modes and adversarial tests

The framework makes failures localizable rather than treating “the model failed” as a single diagnosis.

F1. State-space failure: observed histories cannot be represented in X without violating the declared interpretation.

- F2.** *Preparation failure:* K^P assigns weight to states not producible by the registered procedure, or omits systematic preparation drift.
- F3.** *Dynamics failure:* residuals are structured beyond the declared stochastic and numerical error model.
- F4.** *Measurement failure:* K^Q omits saturation, hysteresis, latency, selection, or calibration uncertainty.
- F5.** *Interface failure:* composition assumes independence or unrestricted coupling where the apparatus enforces correlations or constraints.
- F6.** *Domain failure:* a theorem valid on D is used outside D , or a pointwise limit is reported as uniform.
- F7.** *Equivalence failure:* a formal map preserves equations while changing preparations, boundaries, observables, or referents.
- F8.** *Identifiability failure:* distinct parameter values or models induce the same registered outcome laws.

Lemma 6.1 (Observational quotient). *Exact observational equivalence on fixed D is an equivalence relation on theories sharing the registered outcome spaces.*

Proof. Reflexivity follows from $d_{TV}(P, P) = 0$; symmetry from symmetry of total variation; transitivity follows because if $P_1^c = P_2^c$ and $P_2^c = P_3^c$ for every c , then $P_1^c = P_3^c$ for every c . $\square$

Interpretation 6.2. *The quotient by observational equivalence represents what the registered experiments identify. It should not automatically be read as an ontological quotient. Enlarging D refines or preserves its equivalence classes by restriction monotonicity.*

Definition 6.3 (Registered finite falsifier). *A finite falsifier for T consists of a context c , sample size N , test statistic Z_N , rejection region R , and error guarantees fixed before examining the test data. It rejects the empirical adequacy claim at tolerance $\epsilon(c)$ when $Z_N \in R$ and the test's type-I error is bounded by a declared α under every distribution admitted by that claim.*

Proposition 6.4 (What falsification reaches). *A valid rejection falsifies the conjunction of the model, interpretation, auxiliary calibration assumptions, domain claim, and statistical assumptions used to derive the rejected prediction. It does not logically identify which conjunct failed.*

Proof. The tested distribution is derived from the conjunction. Rejection supplies evidence against that conjunction under the test assumptions. In propositional logic, $\neg(A \wedge B)$ does not entail $\neg A$ and does not entail $\neg B$. $\square$

Empirical statement 6.5 (Minimum evidence record). *A realization or falsification report should contain raw or losslessly transformed observations, context and intervention identifiers, calibration provenance, exclusion rules, uncertainty decomposition, the prediction version, and the prespecified test. Whether a particular record meets these conditions is an empirical and methodological audit question.*

7 Worked finite-dimensional example

Consider the harmonic oscillator

$$X = \mathbb{R}^2, \quad \omega = dq \wedge dp, \quad H(q, p) = \frac{p^2}{2m} + \frac{m\Omega^2 q^2}{2}.$$

Its flow is

$$q(t) = q_0 \cos \Omega t + \frac{p_0}{m\Omega} \sin \Omega t, \quad p(t) = p_0 \cos \Omega t - m\Omega q_0 \sin \Omega t. \quad (7)$$

Assumption 7.1 (Oscillator interpretation). *A preparation device produces a Gaussian $K^P = N((q_0, p_0), \Sigma_P)$; a position sensor has $Y = \mathbb{R}$ and kernel $K^Q(dy \mid q, p) = N(q, \sigma_Q^2)(dy)$; damping and driving are negligible over $0 \leq t \leq T$ within a declared budget.*

Under this assumption, (7) and affine propagation of Gaussian variables yield a Gaussian prediction for each measurement time. That calculation is mathematical. The values of $m, \Omega, \Sigma_P, \sigma_Q$, and the assertion that damping is negligible require calibration and data.

Now perform the canonical rescaling $Q = aq$, $P = p/a$ for $a > 0$. Since $dQ \wedge dP = dq \wedge dp$, this is a symplectomorphism. If preparations are pushed forward and the sensor is transformed to read Q/a , the hypotheses of the empirical-preservation theorem hold. If instead the new sensor is interpreted as reporting Q in the old numerical units, its prediction differs by the factor a . The same canonical transformation is therefore empirically harmless under one interpretation and empirically consequential under another.

For a small neglected quartic correction

$$H_\lambda = H + \lambda q^4,$$

standard continuous dependence of ODE solutions gives convergence to the harmonic flow on compact sets of initial data and compact time intervals as $\lambda \rightarrow 0$. This statement does not supply a global-in-time error bound. An experiment that increases amplitude or duration as λ decreases can expose the correction, mirroring the nonuniform counterexamples above.

8 Consequences for the founding thesis

New result claimed by this work 8.1 (Typed realization criterion). *Within the definitions of this paper, a compositional mathematical structure qualifies as an empirically realized theory only relative to the six-place interpretation data $(\mathcal{C}, \mathcal{P}, \mathcal{Q}, K, \epsilon, D)$ and supporting evidence $\mathcal{E}$. Neither structural admissibility, formal equivalence, nor convergence alone supplies realization.*

Proof. Realization is defined by (2), whose two arguments require both target distributions and interpreted model predictions. Mathematical admissibility specifies neither. Formal equivalence supplies the prediction equality only under the three intertwining conditions. A limiting relation transfers adequacy only with the target-to-family bound in (6). Each omitted ingredient has an explicit counterexample above. $\square$

Conjecture 8.2 (Compositional empirical adequacy). *For a useful class of finite-dimensional classical laboratory systems, empirical interpretations and error budgets can be assigned compositionally so that subsystem adequacy plus verified interface assumptions entails system-level adequacy with nonvacuous bounds.*

This conjecture is stronger than the product error proposition because real interfaces generate correlations, feedback, and model discrepancy. Testing it requires case studies with actual calibration records.

9 Conclusion

The mathematical/empirical boundary can be expressed without weakening mathematical rigor. A model becomes a candidate physical theory only through typed preparation, intervention, measurement, domain, tolerance, and evidence data. Observational equivalence is indexed by available experiments; physical equivalence additionally requires preservation of interpretation and significant structure. Limits require uniformity on the intended domain and an independent empirical bridge. Falsification targets a registered conjunction and must preserve its evidential provenance. These results make the founding thesis precise enough to test, while leaving its general validity conjectural.

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